

EduFlow

A Compliance-Native, Agent-Augmented Operating System for Indian Coaching Centers

Version 1.0 — May 2026

Architecture & Implementation Plan

IIIT Allahabad Founding Team

Submission target: Google XPRIZE — Build with Gemini (Aug 17, 2026)

Prepared for peer review by Gemini, DeepSeek, and collaborating reviewers.

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1. Executive Summary

EduFlow is a **multi-tenant SaaS operating system for Indian coaching centers (10–500 students)** that bundles fee collection, attendance, exams, parent communication, and — critically — **regulatory compliance** (Rajasthan Coaching Act 2025, Central MoE 2024, CCPA, DPDPA 2023, IT Rules 2026).

The product is **agent-augmented**: four production LLM agents (Onboarding, Test Report, Fee Collection / Smart Dunning, Compliance Watch) handle high-leverage operational work. Most flows are deterministic workflows per Anthropic's December 2024 taxonomy; only open-ended parent/student interactions use autonomous-agent loops.

The wedge is compliance. Competitors (Classplus, Teachmint, Proctur) sell commodity LMS+CRM at ₹8K–50K/year. None ship Rajasthan-Act-ready registration tracking, refund pro-rata enforcement, CCPA-safe ad lint, or DPDPA verifiable parental consent. With the CCPA already imposing ₹1.39 Cr in penalties on 31 coaching institutes (PIB Nov 2024) and Rajasthan's ₹2–5 lakh per-violation regime live since September 2025, this is a defensible, recurring, insurance-style sale.

Pricing: ₹999 / ₹2,499 / ₹4,999 per month (Starter / Pro / Compliance+), targeting ₹50L–1 Cr ARR ceiling over 24 months.

Bootstrap stance: ₹5L of family/student capital, no VC, founders self-allocate full-time, AI COGS funded by Sarvam + Anthropic startup-program credits in year 1.

This document combines findings from five parallel research streams (educational AI agents, multi-tenant SaaS + LangGraph orchestration, Indian AI vendors, DPDPA/regulatory, real coaching-center operations) into a single, peer-reviewable architecture and execution plan.

2. Problem Statement & Market Context

2.1 Market size and segmentation

- **Total Indian coaching market:** USD 7.2B in 2025 → USD 17.8B by 2034, 10.29% CAGR (IMARC).
- **NSS 2025 data:** 27% of all Indian students take private coaching (30.7% urban, 25.5% rural). Urban higher-secondary spend is ₹9,950/student/year.
- **Target segment:** small/mid coaching centers (10–500 students) in tier-2/3 cities (Prayagraj, Agra, Kota, Lucknow, Jaipur, Bhopal, Indore, Patna, Ranchi, Bhubaneswar).
- **Estimated SAM:** ~120,000–180,000 such centers nationally; capture target = 1,000 paying customers within 24 months (~0.6% penetration).

2.2 The day-in-the-life of a coaching owner

A 100–300 student coaching owner works ~12 hours/day. Time allocation:

Activity	Hours/day
Teaching	3–4
Fee chasing & parent calls	2–3
Scheduling, substitutes, faculty issues	2
Test/material prep	1
Walk-ins, inquiries, conversion	1
Admin (spreadsheets, registers)	1

Industry workflow vendors estimate ~2 days/week reclaimable through automation of fees, attendance, and parent comms.

2.3 Real pain points (ranked by willingness-to-pay)

Rank	Pain	Quantified loss	WTP/month
1	Late / leaking fee collection	20–40% of fees >30 days late at baseline; ₹30K–1L locked at any time on a 200-student center	₹1,500–3,000
2	Compliance / inspector / refund risk	₹2 lakh per Rajasthan violation, ₹50 lakh per CCPA misleading-ad notice	₹2,000–5,000
3	Faculty churn → student loss	25% sector attrition; FIITJEE-style mass exits can wipe out a batch	₹1,500–3,000
4	Slow inquiry response (lost enrollments)	5-minute response = 21× conversion lift (Reshape)	₹1,500–2,500
5	Student dropout / parent anxiety	5% retention gain on 200 students = ₹1L+/month	₹1,000–2,000

Total wallet for a 200-student tier-2 center: ₹6,000–15,000 / month — 3–5× commodity ERP price because the features pay for themselves in measurable rupees.

2.4 Why this market is hard

- **Meta WhatsApp Business AI launched in India on May 7, 2026** — free, multilingual, with UPI payments coming. It commoditizes generic FAQ chat, lead capture, appointment booking. Any coaching SaaS whose wedge is "AI-powered WhatsApp reminders" is dead on arrival.
- **Classplus pivoted away from B2B SaaS** to test-prep content (Testbook acquisition) because pure-SaaS unit economics didn't hold (Inc42).
- **Indian edtech funding hit an 8-year low** (–56% YoY to \$249M in 2025) — no VC route for another B2B coaching SaaS.
- **Teachmint cut headcount 662 → 187** and still couldn't crack profitable scale.

The viable path threads the needle: bootstrap, ₹999–4,999/month, compliance-led wedge, founder-led Hindi GTM, integrate **alongside** WhatsApp Business AI rather than competing with it.

2.5 Compliance landscape (this is the wedge)

Regulation	Effect	Penalty
Rajasthan Coaching Centres Act 2025 (Sep 2025)	Registration mandate for 100+ student centers, 1 sqm/student, max 5 hrs/day, ≥4-instalment fees, 10-day pro-rata refund, mental-health counsellor, no misleading ads	₹2L / ₹5L / cancellation
MoE Central Guidelines 2024	No enrolment <16, tutor must be graduate, no guarantee-rank ads, refund-policy disclosure	State-enforced
CCPA Coaching Sector Guidelines 2024	No-refund clauses per-se invalid; selection-claim must have evidence; banned terms list	₹10L–50L per offence; ₹1.39 Cr collected on 31 institutes already (PIB)
DPDPA 2023 + Rules (notified Nov 2025, enforcement May 2027)	Verifiable parental consent for all <18 students (no COPPA-style 13/16 carve-out); 90-day rights SLA; 72-hour breach notification	₹250 Cr (security safeguards), ₹200 Cr (child data), ₹50 Cr (catch-all)
IT Rules 2026 (Feb 10 2026)	All AI-generated content must carry visible watermark + tamper-resistant provenance; 3-hour takedown SLA	Intermediary liability

A coaching SaaS that ships these as features is a non-bypassable insurance product.

3. Product Positioning & Wedge

3.1 One-line positioning

"The compliance-native operating system for Indian coaching centers. Classplus gives you an app. EduFlow runs your coaching — and keeps you out of court."

3.2 What we build, what we don't

Build (core 5 modules):

1. **Smart Dunning Engine** — fee collection automation with behavioural-econ-tuned cadences, WhatsApp utility templates, UPI deep links.

2. **AI Test & Grading** — curriculum-tagged question bank, one-click shuffled PDFs, OMR + handwritten Devanagari grading, chapter-wise analytics.
3. **ComplyShield** — Rajasthan/UP/Karnataka/Bihar/Central rules engine, registration filings, refund calculator, ad-copy lint, inspector-ready PDF pack.
4. **Faculty & Batch Margin Analytics** — per-batch revenue attribution, faculty cost, gross margin per teacher, student rating, retention forecast.
5. **Inquiry CRM + Parent Digest** — 60-second auto-greet on WhatsApp leads, demo scheduler, referral attribution; weekly auto-digest to parents with attendance + test scores + one small-win highlight.

Don't build (commoditized or low-value):

- Generic LMS / live class streaming (Zoom + Google Meet own this; PhysicsWallah dominates free side)
- Branded mobile app per coaching (Classplus race-to-bottom; parents won't install N apps)
- Yet-another student quiz app (Testbook/Embibe/Unacademy saturate this)
- GST/accounting (Tally/Zoho own it; integrate, don't compete)
- Student social feed / forum (engagement low, moderation cost high, suicide-prevention liability)

3.3 Pricing tiers

Tier	Monthly	Includes	Target customer
Starter	₹999	Smart Dunning + Inquiry CRM + Parent Digest, up to 50 students	Single-owner tier-3 coaching
Pro	₹2,499	+ AI Test & Grading + Faculty Analytics + Risk Score, up to 200 students	Mid tier-2 coaching
Compliance+	₹4,999	+ ComplyShield (state-specific) + audit pack + DPDPA consent broker, up to 500 students	Regulated tier-1/2 coaching
Per-student add-on	₹15 / student / month	Alternative to flat plan; same features	Centers that prefer usage pricing

OCR-heavy grading metered separately at ₹3/page beyond 100/month in Pro, 200/month in Compliance+.

3.4 What positioning rejects

This positioning explicitly **abandons** the "AI school management" framing. AI is the *infrastructure*, not the *product*. The product is operational outcomes (fees collected, students retained, inspectors satisfied, faculty rated). The XPRIZE judges will reward genuine AI-native operations — but we sell to the owner on rupees.

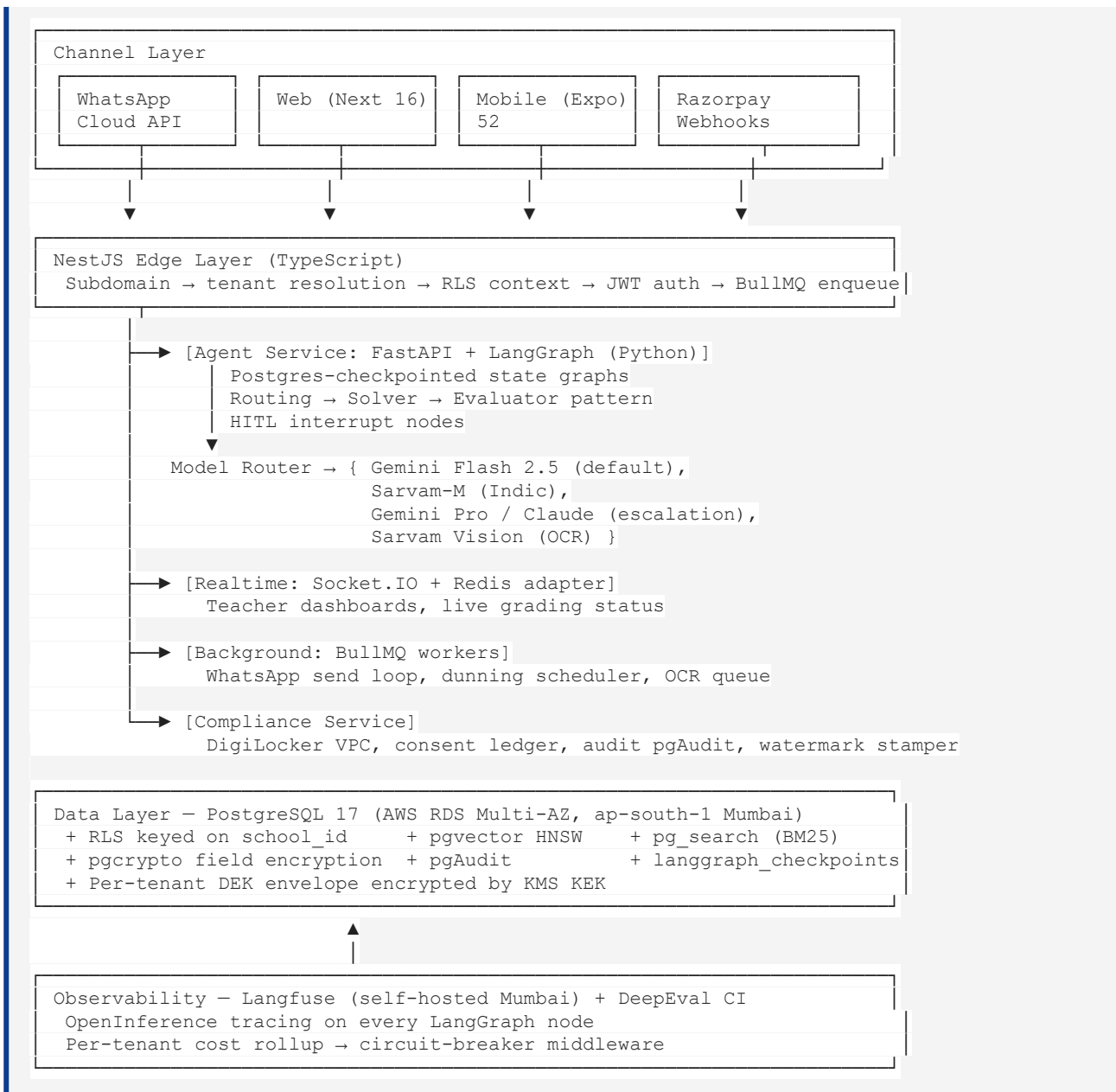
4. System Architecture

4.1 Architectural philosophy

Five principles derived from the research:

1. **Workflows over agents.** Per Anthropic's "Building Effective Agents" (Dec 2024), most production tasks should be structured workflows (routing, prompt chaining, orchestrator-workers, evaluator-optimizer). Autonomous agent loops are reserved for genuinely open-ended interactions.
2. **Human-in-the-loop at every write boundary.** IT Revolution data shows autonomous pipelines have a 23% critical-error rate without HITL gates; structured gates drop this to 5.1%. EduFlow gates: any fee-structure write, any escalated dunning message, any compliance document send.
3. **Crypto-shred for right-to-erasure.** Per-tenant Data Encryption Keys (DEKs) envelope-encrypted by a KMS Key Encryption Key (KEK). Right-to-erasure = drop the DEK; ciphertext becomes unreadable. Avoids cascading delete races at scale.
4. **Postgres-first.** Single source of truth for transactional state, RLS for tenancy, pgvector + pg_search for hybrid retrieval, PostgresSaver for LangGraph checkpoints. Avoid multi-DB ops complexity at <1000 customers.
5. **India-region everything.** Postgres in AWS ap-south-1 (Mumbai), Langfuse self-hosted in Mumbai, default LLM = Azure OpenAI South India or Gemini Vertex Mumbai. PII anonymizer gateway before any non-India inference call.

4.2 Reference architecture (textual diagram)



4.3 Component-by-component

4.3.1 Tenancy

- **Single PostgreSQL cluster**, all tenant-scoped tables include `school_id uuid not null`.
- **Row-Level Security**: `FORCE ROW LEVEL SECURITY` keyed on `current_setting('app.current_tenant_id')`.
- **Prisma extension** (`row-level-security` pattern) opens a transaction per request, calls `SELECT set_config('app.current_tenant_id', $1, true)`, runs query, commits. Transaction-scoped config is PgBouncer-safe.
- **PgBouncer** in transaction-pool mode in front of RDS.

- **Composite indexes** lead with `school_id`; analytics role has `BYPASSRLS` for internal dashboards.
- **Schema-per-tenant rejected**: PlanetScale's 2025 analysis shows shared system catalogs slow the planner past a few hundred tenants. Database-per-tenant reserved as enterprise upsell for districts that demand it.

4.3.2 NestJS edge

- Three logical modules: `core` (tenancy, auth, billing), `agent` (LangGraph client), `channels` (WhatsApp webhook, Razorpay webhook, Socket.IO).
- Middleware order: tenant-from-subdomain → JWT verify → RLS context set → handler.
- Async-Local Storage for per-request context (`tenant_id`, `user_id`, `consent_id`) so it propagates to all internal calls.

4.3.3 Agent service (LangGraph)

- Separate FastAPI service, Python 3.12, deployed alongside NestJS on ECS Fargate.
- Called from NestJS via internal gRPC.
- `PostgresSaver` checkpoint in a separate `langgraph_checkpoints` schema in the same Postgres — gives durable execution across multi-day flows like dunning cadences.
- **State always carries** `{tenant_id, user_id, consent_id, conversation_id}` — propagated through every node.
- `interrupt()` nodes for HITL approval gates.
- `Send` API for fan-out so checkpoint consistency holds across parallel branches.

4.3.4 Realtime

- Socket.IO server in the NestJS process, Redis adapter for horizontal scaling.
- Used for teacher dashboards (live grading status, attendance check-in), owner dashboards (today-view live updates).

4.3.5 Background workers

- BullMQ on Redis (ElastiCache).
- **Queues**: `whatsapp_inbound`, `whatsapp_outbound`, `dunning_scheduler`, `ocr_grading`, `nightly_digest`, `compliance_renewal_check`.
- Rate-limited per-tenant against Meta's tiered messaging limits.

4.3.6 Web & mobile

- **Next.js 16** App Router on Vercel with multi-tenant routing (Vercel for Platforms pattern: `acme.eduflow.in`).

- **Expo 52** mobile shell sharing API contract.
- Same auth (JWT + refresh in httpOnly cookie web, Expo SecureStore mobile).
- Tailwind v4, shadcn/ui, React Server Components for owner dashboard.

4.3.7 Payments

- **Razorpay Subscriptions** for school plans (one Plan per tier, one Subscription per school).
- UPI eMandate primary, cards secondary.
- Webhook on `subscription_charged` → updates `subscriptions` table → entitlements refreshed in JWT claims at next login.
- Per-student fee collection handled outside the SaaS plan layer — coaching center owns the Razorpay account; EduFlow generates payment links on their behalf.

4.3.8 Deployment

- **AWS ap-south-1 (Mumbai)** primary.
- **2-AZ Multi-AZ** for RDS, ElastiCache, ECS.
- **ECS Fargate** for NestJS + FastAPI services behind ALB.
- **S3 Object Lock (Compliance Mode)** for audit log backups and student artifacts (test PDFs, answer sheets, refund proofs).
- **CloudWatch + Langfuse + Sentry** for infra / LLM / app errors.
- **Vercel** for marketing + web app.
- **EAS Build** for Expo mobile artifacts.

5. AI Agent Specifications

5.1 Design pattern selection

Following Anthropic's "Building Effective Agents":

Agent	Pattern	Rationale
Onboarding	Routing → Orchestrator-Workers	Setup steps are fixed, but each step's content depends on coaching's profile.

Test Report	Evaluator-Optimizer	Generator drafts a report; evaluator scores against the 8 Maurya pedagogical dimensions; loop ≤ 3 iterations.
Smart Dunning	Workflow with HITL interrupt	Cadence is deterministic; stage 4+ requires owner approval.
Compliance Watch	Routing $\rightarrow$ Tool-Use	Match rule changes to coaching's profile, generate filings.
Doubt Resolution (V2)	Autonomous Agent	True open-ended; tool path unknown until reasoning.

5.2 Onboarding Agent

Goal: Convert a fresh coaching-center signup into a fully configured tenant within 24 hours via WhatsApp in Hindi/Hinglish.

Inputs: Owner phone, school name, city, estimated student count, screenshot or CSV of existing batch/fee data.

Outputs: - Created `school`, `batches`, `students`, `fee_structures` records - DPDPA-compliant privacy notice URL sent to owner - Day-1 demo class scheduled with the owner

Flow: 1. Welcome message in Hindi (via WhatsApp template, utility category). 2. Ask for school name $\rightarrow$ confirm via menu. 3. Ask for batches (e.g., "Class 10 Maths Morning", "JEE Foundation Evening"). 4. Ask for fee structure per batch. 5. (Optional) Owner uploads spreadsheet/screenshot of existing students $\rightarrow$ Sarvam Vision OCR $\rightarrow$ LLM normalizes to canonical JSON $\rightarrow$ owner confirms. 6. Generate per-student parent consent URLs (DigiLocker flow). 7. Send owner a setup summary + dashboard link.

Model routing: - Router (intent classification): Gemini Flash 2.5 - Hindi conversation: Sarvam-M - Spreadsheet/screenshot OCR: Sarvam Vision - Data normalization (JSON extraction): Gemini Flash 2.5 with strict JSON schema

HITL gates: - After OCR ingestion: owner must confirm parsed student list before write. - Before activating fee structure: owner approves.

Eval metric: Time-to-first-active-batch ≤ 24 hours; CSAT $> 4.0/5$ from owners.

5.3 Test Report Agent

Goal: Generate a personalized per-student progress report after a test, with weak topics, comparative analytics, and study plan.

Inputs: Test ID, raw scores (per student, per question), curriculum tags, optional handwritten answer-sheet scan.

Outputs: - Per-student PDF report (with C2PA watermark) - WhatsApp-friendly text summary for parent - Updated Bayesian Knowledge Tracing scores in `student_kt` table

Flow: 1. **Deterministic compute** (Python tool, not LLM): aggregate scores, compute percentile rank, chapter-wise mastery, identify top-3 weak topics. This avoids LLM math hallucination per the Khanmigo / Kestin RCT lessons. 2. **Update KT scores:** Beta(α , β) per concept; do not let LLM hallucinate mastery. 3. **RAG retrieval:** pull 3–5 curriculum-aligned worked examples for the weak topics from the embedded curriculum corpus. 4. **Generator** (Gemini Flash 2.5): drafts the report using the 8 Maurya pedagogical dimensions (mistake ID, mistake location, guidance, actionability, tone, accuracy, age-appropriateness, clarity). 5. **Evaluator** (Gemini Flash 2.5 as judge, non-deterministic temp 0.7): scores draft against 8 dimensions. 6. If any dimension < 7/10, regenerate (≤ 2 retries). 7. **Translate** to Hindi via Sarvam Translate. 8. **Watermark + provenance:** PDF gets C2PA metadata + visible footer "AI-assisted content. Generated by EduFlow on . Verify before action." 9. **Teacher approval** (HITL): teacher dashboard shows draft; one-tap approve → send to parent.

Why this design wins: - Deterministic compute layer means scores are always correct. - RAG layer means content is curriculum-anchored, not hallucinated. - Eval-optimizer loop catches obvious failures before teacher review. - Teacher HITL means hallucination liability stays with teacher's judgement, not pure AI.

Eval metric: Teacher edit rate < 20% (acceptance with minor tweaks); parent-rated usefulness > 4.0/5.

5.4 Smart Dunning Agent

Goal: Recover overdue fees through a behaviourally-tuned cadence in Hindi/Hinglish, while protecting the coaching center's parent relationships.

Cadence (grounded in PNAS 2024 behavioural-econ, Dutch field experiment 23.4% lift, ACF BIAS report):

Stage	Timing	Channel	Tone	Auto/HITL
1	T-3 days	WhatsApp utility template	Friendly nudge, percentage framing, payment link	Auto
2	T+0	WhatsApp utility template	Due-today reminder, social proof ("85% of parents have paid")	Auto
3	T+3	WhatsApp utility template	Faculty-signed, single follow-up	Auto
4	T+7	Phone call script + WhatsApp	Owner-level, no auto-send	HITL — owner must trigger
5	T+14	Formal letter	Escalation to legal-tone	HITL — owner must approve text

Guardrails (RBI FREE-AI principles, FDCEPA-equivalent norms):

- **Hard cap:** max 1 reminder per day, max 4 per week per parent.
- **Quiet hours:** 21:00–09:00 IST, no sends.
- **Stop-words:** parent says "stop", "lawyer", "court" → escalate to human owner, never auto-reply.
- **Mandatory disclosure:** every message labelled "Sent on behalf of via EduFlow (AI-assisted)".
- **Full audit log** of every send (DPDPA + RBI requirement).

Personalization: - Segment by (days overdue, fee size, payment-latency history, parent vs student account). - Tone calibration: Indian middle-class face-saving sensitivity — never threaten; always offer instalment renegotiation in stages 3+.

Model routing: - Template fill (utility category): Gemini Flash 2.5 with strict JSON schema. - Parent reply analysis: Sarvam-M (Hinglish handling). - Escalation tone (stage 5): Claude Opus 4.7 via startup-program credits.

Eval metric: Days-to-collect reduction > 20% vs baseline; opt-out rate < 2%; zero complaints to coaching owner.

5.5 Compliance Watch Agent (ComplyShield)

Goal: Keep the coaching center continuously compliant with Rajasthan / UP / Karnataka / Bihar / Central rules; generate inspector-ready filings on demand.

Watch sources: - State coaching-act gazette feeds (scraped weekly) - CCPA notification page - MoE press releases - Sector news (ThePrint, Business Standard) for enforcement actions

Capabilities: 1. **Registration & renewal:** state-aware wizard, auto-generated PDF application, renewal alerts at T-90 / 60 / 30 days. 2. **Infrastructure tracker:** 1 sqm/student calculator, batch-size cap, CCTV uptime monitor (Hikvision/CP+ RTSP integration), first-aid checklist with monthly attestation. 3. **Fee & refund compliance:** ≥4-instalment enforcement, pro-rata refund calculator (Rajasthan 10-day rule), UPI payout generator, audit trail. 4. **Ad-copy lint:** pre-publish CCPA scanner; flags "100% selection", "guaranteed rank", false scarcity, unverified selection claims. 5. **Faculty compliance:** graduation-degree upload, background-check log, contract template library. 6. **Mental-health duty of care:** counsellor assignment per 200 students, Student Risk Score → counsellor escalation, helpline footer on parent messages, quarterly auto-PDF report. 7. **Inspector Pack:** one-click PDF bundle of all of the above with tamper-evident hash chain. 8. **Penalty risk dashboard:** real-time score "Your Rajasthan compliance is 87% — 3 issues could trigger ₹2L penalty" with one-click remediation flow.

Model routing: - Rule-change ingestion: Gemini Flash 2.5 with JSON extraction. - Filing generation: Gemini Pro (longer reasoning). - Ad-copy lint: lightweight classifier + Sarvam-M for Hinglish.

HITL gates: every filing generation requires owner approval before submission to state authority.

Eval metric: Customer-reported penalty avoidance; inspector-pack completeness > 95%.

5.6 Auxiliary agents (V1.5 / V2)

- **Student Risk Score:** weighs attendance (40%), test trend (30%), fee delinquency (15%), parent contact recency (15%). Threshold breach → counsellor outreach.

- **Parent Weekly Digest:** auto-summarizes attendance, last 2 test scores with peer average + topper benchmark, one teacher-tagged "small win", upcoming test date. Sent via WhatsApp utility template.
- **Inquiry CRM:** 60-second auto-greet on WhatsApp leads from Meta/Google ads, demo scheduler, referral attribution. Single biggest conversion-lift feature in the stack per Reshape data (21× lift on 5-min response).
- **Doubt Resolution (V2):** true autonomous agent for student questions; reserved for future once data shows demand.

6. Data Model & Multi-Tenancy

6.1 Core entities

```
schools (tenant root)
├─ users (owner, teacher, counsellor, admin)
├─ batches
│   └─ students
│       ├── guardians (parent records with VPC token)
│       ├── enrolments
│       ├── attendances (partitioned monthly)
│       ├── test_attempts
│       │   └─ question_responses
│       ├── payments
│       ├── fee_structures
│       ├── dunning_events (audit)
│       └─ kt_state (Bayesian KT scores per concept)
├─ faculties
│   ├── contracts
│   ├── batch_assignments
│   └─ ratings
├─ inquiries
│   ├── touchpoints
│   └─ conversions
├─ compliance_records
│   ├── registrations
│   ├── refunds
│   ├── ad_lints
│   └─ counsellor_sessions
├─ ai_generations (provenance ledger)
├─ consent_events (append-only)
└─ audit_log (append-only, monthly partitioned)
```

6.2 Multi-tenancy implementation

Every tenant-scoped table has:

```
school_id UUID NOT NULL REFERENCES schools(id),
-- Composite index leads with school_id
INDEX idx_<table>_school_id_<col> (school_id, <other_col>)

-- RLS policy
ALTER TABLE <table> ENABLE ROW LEVEL SECURITY;
ALTER TABLE <table> FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON <table>
```

```
USING (school_id = current_setting('app.current_tenant_id')::uuid);
```

Prisma extension wraps every request:

```
const tenantExtension = Prisma.defineExtension({
  query: {
    $allOperations: async ({ args, query, model, operation }) => {
      const tenantId = AsyncLocalStorage.get('tenant_id');
      return prisma.$transaction(async (tx) => {
        await tx.$executeRawUnsafe(
          `SELECT set_config('app.current_tenant_id', '${tenantId}', true)`
        );
        return query(args);
      });
    },
  },
});
```

6.3 Encryption strategy

Three tiers:

1. **Storage encryption** (infra): AWS RDS at-rest with KMS CMK in ap-south-1 — covers compliance baseline, transparent to app.
2. **Field-level pgcrypto** (app): AES-256-GCM on `students.dob`, `guardians.phone`, `guardians.aadhaar_hash`, `students.exam_scores`, `payments.upi_id`. Keys in AWS KMS, rotated yearly.
3. **Per-tenant DEK** (crypto-shred): each tenant has a Data Encryption Key, envelope-encrypted by a KMS Key Encryption Key. Right-to-erasure = drop the DEK; ciphertext becomes garbage. No cascading delete races.

6.4 Append-only ledgers

`consent_events`, `audit_log`, `ai_generations`, `dunning_events` are append-only:

- No UPDATE, no DELETE allowed at the database role level for application user.
- Monthly partitions for `audit_log` and `attendances`.
- Hourly export to S3 Object Lock (Compliance Mode) for immutable backup.
- `pgaudit` extension enabled with `log = 'write, ddl'`.

6.5 Vector & search

- `pgvector` 0.8 with HNSW index for embeddings.
- `pg_search` (ParadeDB) for BM25 in the same Postgres.
- Curriculum corpus chunked via `MarkdownHeaderTextSplitter` (textbook structure) → recursive 512/50 overlap fallback.

- Hybrid retrieval: BM25 top-20 □ vector top-20 → Reciprocal Rank Fusion → Cohere Rerank-3 → top-5 → LLM. Two-stage gives +17% Recall@5 over single-stage (OptyxStack 2025).
- **Embedding model:** Vyakyarth-1 (Krutrim, Indic-contrastive) for Hindi/Indic; Cohere embed-multilingual-v3 as managed fallback for mixed content.
- **Privacy:** opaque IDs in vector metadata; never store student names. For high-risk content (essays), add Gaussian noise (DP, $\epsilon=1.0$) to embeddings to mitigate inversion attacks (arXiv 2411.05034).

7. DPDPA + IT Rules 2026 Compliance Architecture

7.1 Layered compliance stack

```

Layer 7 - Governance: DPIA register, data_map.yaml in repo, audit log
Layer 6 - Rights Portal: /privacy with access, correction, erasure, portability (90-day SLA)
Layer 5 - Consent Layer: DigiLocker VPC + Consent Manager SDK (post Nov 2026)
Layer 4 - AI Gateway: PII anonymiser → LLM router → watermark + provenance ledger
Layer 3 - Application: child-flag aware, minor-privacy-switch, RBAC, MFA enforced
Layer 2 - Data Plane: Postgres + pg_tde + pgcrypto + per-tenant DEK + Mumbai region
Layer 1 - Infra: AWS ap-south-1, VPC isolation, CloudTrail immutable, WAF, Vault secrets

```

7.2 Verifiable Parental Consent (VPC) flow

Primary path: DigiLocker OAuth

1. Student signs up; age check → if <18, account stays in `pending_consent` state.
2. Student provides parent's phone number.
3. EduFlow sends WhatsApp message to parent with DigiLocker authorisation link.
4. Parent authenticates via Aadhaar-linked DigiLocker; platform receives a signed verified-adult token.
5. Token, parent identity hash, child_id, purpose scope written to `consent_events` (append-only).
6. Student account activated.

Fallback path: Phone OTP + Aadhaar XML offline-KYC for age verification, then PAN/Aadhaar Offline KYC, then OTP-signed consent.

Withdrawal: parent visits `/privacy` (or sends WhatsApp `STOP` to a dedicated number) → consent withdrawn → next data access blocked → 90-day deletion timer starts.

Re-consent triggers: any change in data-use purpose; new AI feature added that processes student data.

7.3 IT Rules 2026 — Synthetic content watermarking

Every AI-generated artefact (test reports, parent messages, study plans, dunning copy) carries:

1. **Visible watermark:** PDF footer "Generated by EduFlow AI on . AI-assisted content."
2. **C2PA provenance metadata** embedded in the file.
3. **Audio TTS:** prepended with "The following is AI-generated content."
4. **Provenance ledger:** `ai_generations` table with `(id, model, prompt_hash, output_hash, student_id, tenant_id, timestamp, channel)`.
5. **3-hour takedown SLA:** `/grievance` endpoint auto-pages on-call; SLA tracker enforces 3 hours.

7.4 Breach playbook

- **Detection:** CloudTrail anomaly + Sentry alerts + database access pattern monitor.
- **72-hour clock starts** at first credible indicator of breach.
- **Notify:** Data Protection Board + affected Data Principals via email + dashboard banner.
- **Template notices** pre-drafted in repo (`/runbooks/breach-notification/`).
- **On-call rotation** with weekly drill.

7.5 10-item MVP compliance checklist (ship by Sept 2026)

1. Age-gate + DigiLocker VPC on every signup.
2. RDS storage encryption + pgcrypto field encryption + per-tenant DEK.
3. Privacy notice in Hindi + English + 4 regional languages.
4. Rights portal at `/privacy` with 90-day SLA timer.
5. Published grievance contact `dpo@eduflow.in`.
6. AI gateway: PII anonymizer + default to Azure OpenAI South India; provenance ledger; visible watermark.
7. MFA on all admin / teacher / support accounts.
8. 72-hour breach playbook with on-call rotation.
9. `data_map.yaml` reviewed per PR that touches schema.
10. Behavioural-monitoring kill-switch default-on for minor accounts.

7.6 Post-launch compliance roadmap (2026–2027)

1. Engage MeitY-registered Consent Manager (Leegality / Consently / Digital Anumati) once registration list publishes (Nov 2026).
2. Annual DPIA + independent audit; DPO-as-a-Service (₹15–50K/month).

3. Cyber insurance ₹10 Cr cover with regulatory-penalty rider (Marsh / ICICI Lombard / HDFC Ergo).
4. Migrate Indic inference fully to Sarvam / AI4Bharat self-hosted in ap-south-1 as SDF localisation list firms up.
5. Differential-privacy analytics + federated fine-tuning post 20-customer threshold.

8. Indic Language Stack

8.1 Model routing matrix

Task	Model	Why
Hindi/Hinglish parent message draft	Sarvam-M	+86% on romanized GSM-8K; best native Hindi tone
Fee reminder template fill (mixed English/Hindi)	Gemini 2.5 Flash	Cheapest reliable structured-output
Test report generation (English reasoning)	Gemini 2.5 Flash	Reasoning quality; cost
Final Hindi polish of report	Sarvam Translate	Production-grade Indic translation
Handwritten Devanagari answer-sheet OCR	Sarvam Vision	95.91% Hindi doc accuracy; beats Gemini 3 Pro by 4 pts
Mixed English-math answer sheet	Gemini 2.5 Pro vision	Better on math + figures
Customer onboarding voice call	Sarvam Saaras v3 + Sarvam-M	Indic-native voice loop
Curriculum embeddings	Vyakyarth-1 / Cohere embed-multilingual-v3	Indic-tuned vs managed quality
Complex parent dispute / refund negotiation	Claude Opus 4.7 (startup credits)	Premium reasoning when stakes high
Real-time WhatsApp reply	Gemini 2.5 Flash + prompt cache	Latency + cost

8.2 Anonymizer gateway

Before any cross-border LLM call:

1. Regex + NER strips: name, phone, email, school name, exam-id, address.

2. Replaces with `[STUDENT_001]`, `[GUARDIAN_001]` opaque tokens.
3. Tokens persisted in `pii_tokens` (short-TTL, tenant-scoped).
4. After response, tokens rehydrated to original PII.
5. Every cross-border call logged with payload hash + token map hash.

For child-data flows, default-route to Azure OpenAI South India or Sarvam (India region); only use Anthropic / OpenAI (US) when explicit re-consent recorded for that feature.

8.3 Cost estimates

Per-customer per-month at 1000 customers (50 WhatsApp utility msgs, 200 LLM-generated msgs, 5 voice min, 100 OCR pages, 30 embedding ops):

Line	Cost
Gemini 2.5 Flash	~\$1.86
Sarvam-M	~₹30
Sarvam Vision OCR	~₹200 (passed through as ₹3/page beyond plan limit)
Sarvam STT/TTS	~₹2.5
Sarvam Translate	~₹20
Cohere Embed	~\$0.15
WhatsApp via Cloud API	~₹30
Vertex AI infra	~\$0.20
Claude premium (5% calls, post-credits)	~\$0.40
Total ≈ ₹560 (\$6.70) per customer per month	

At ₹999 Starter pricing: 44% gross margin. At ₹2,499 Pro: 78% gross margin. Both sustainable.

Year 1: Sarvam Startup Program + Anthropic Startup Program credits cover ~50% of variable cost.

9. Observability & Evaluation

9.1 Tracing stack

- **Langfuse self-hosted** in AWS ap-south-1 (DPDPA-compliant; no student traces to US SaaS).
- **OpenInference instrumentation** auto-captures LangGraph nodes as spans.
- Each span carries `tenant_id`, `user_id`, `consent_id` (which DPDPA grant covered this call), prompt hash, output hash, model, tokens-in/out, cost, eval verdict.
- **Per-tenant cost rollup** in Postgres feeds a circuit-breaker middleware: when tenant approaches monthly LLM budget, downgrade to cheaper model or queue.

9.2 Three-layer evaluation pipeline

Every prompt / model change runs through:

Offline pedagogical eval (DeepEval in CI): - 200+ anonymized real student situations - LLM-as-judge (Gemini 2.5 Pro, temp 0.7, no CoT) scoring against 8 Maurya dimensions - Calibrated against 50 human-graded samples quarterly

Bias eval, stratified: - Paired test set varying only student name (Hindu/Muslim/Dalit-coded) or Hinglish script proficiency - Measure tone, recommended action, report-length deltas - Blocking failures: > 5% delta in recommended action

Adversarial / prompt-injection eval: - OWASP LLM Top 10 suite - Student-data-extraction attacks (arXiv 2512.03694 method) - Trojanized-prompt-chain detection (arXiv 2507.14207)

9.3 Agent-specific metrics (DeepEval)

- **Goal Accuracy:** did the agent achieve the user's intent?
- **Plan Adherence:** did the agent follow the expected control flow?
- **Tool Use:** were tools called correctly?
- **Step Efficiency:** minimal steps to completion?

Tracked per tenant, surfaced in customer health dashboard.

10. Implementation Plan (90-day shipping)

10.1 Sprint structure

Six 2-week sprints from May 28 to Sept 4, 2026, ending with XPRIZE submission packaging Sept 5–17.

Sprint 1 — May 28 to Jun 10: Foundation + Pilot Zero

Engineering: - Repo rebrand: rename modules, kill school-specific copy, update README - Strip non-coaching modules: Class Wall, Chat, Homework (Google Classroom replaces these for free) - Add `language` column to all user-facing tables; default `hi` - Set up AWS ap-south-1: RDS PostgreSQL 17, ElastiCache Redis, S3 buckets, KMS keys - Apply Prisma RLS extension across all tenant-scoped models - Add `pgaudit`, `pgvector`, `pg_search`, `pgcrypto` extensions - Deploy NestJS to ECS Fargate behind ALB - Set up Langfuse self-hosted in ap-south-1 - Compliance skeleton module: `consent_events`, `audit_log`, `ai_generations` tables

GTM: - Founder 1 (full-time): build list of 20 coaching owner contacts via family, IIITA alumni, hometown network - 10 Hindi phone calls; pitch concierge pilot (free 30-day + ₹599/mo after) - **Target: 1 verbal yes**

Setup: - GitHub Student Pack, Google Cloud \$300, NVIDIA Inception, Microsoft for Founders Hub apply - Udyam + Razorpay KYC - Apply Sarvam Startup Program day 1 - Apply Anthropic Claude Startup Program (need IIITA prof advisor) - Buy `eduflow.in` domain

Content: - YouTube channel launch - Video #1 in Hindi: "Coaching center kaise efficiently chalaye — 5 mistakes you're making"

Sprint 2 — Jun 11 to Jun 24: Smart Dunning + Onboarding Agent

Engineering: - WhatsApp Cloud API integration: NestJS webhook controller with HMAC verify; BullMQ workers; utility template submission to Meta for approval (target 6 templates) - LangGraph Python service deployed on ECS, FastAPI exposing gRPC endpoint - Onboarding Agent (Sprint 1 spec): - State graph: welcome → school_name → batches → fee_structure → student_upload → confirm - Sarvam-M for conversation, Sarvam Vision for spreadsheet OCR - HITL confirm node before write - Smart Dunning Agent (Sprint 1 spec): - 5-stage cadence with WhatsApp utility templates - Behavioural-econ-tuned copy in Hindi + English - Hard caps + quiet hours + stop-word handling - Audit log of every send

GTM: - Onboard pilot zero — 2 days in coaching center if hometown, else full Zoom - Founder 1 conducts onboarding personally; record interactions for product feedback - Founder 5 captures testimonial video (30 sec Hindi phone-shot)

Content: - Video #2: "Rajasthan Coaching Act ka asar — chhote coachings ke liye" - Set up Click-to-WhatsApp Meta ads account (configure, don't launch yet)

Sprint 3 — Jun 25 to Jul 8: Test Report + AI Grading

Engineering: - Curriculum corpus ingestion: NCERT + CBSE Class 9-12 + JEE Foundation + NEET Foundation - Chunked via MarkdownHeaderTextSplitter - Embedded via Vyakarth-1 (Indic) + Cohere embed v3 (mixed) - pgvector HNSW + pg_search BM25 hybrid index - Test Report Agent: - Deterministic compute layer (Python tool): score aggregation, percentile, chapter mastery - Bayesian Knowledge Tracing: `student_kt` table with Beta(α , β) per concept - Generator (Gemini Flash 2.5) → Evaluator (Gemini Flash 2.5 judge) → ≤ 3 retries - Hindi polish via Sarvam Translate - C2PA watermark + provenance ledger - Teacher HITL approval node - OCR module: Sarvam Vision for handwritten Devanagari; Gemini 2.5 Pro vision fallback for math - DPDPA: rights portal `/privacy` skeleton with access + correction + erasure forms

GTM: - Launch Click-to-WhatsApp Meta ads — ₹500/day, target Prayagraj/Agra/Kota/Lucknow coaching owners - Founder 1 conducts Hindi Zoom demos; founder 5 records for content reuse - **Target: 1 paying customer + 3-5 active free trials**

Content: - Video #3: "EduFlow product walkthrough — Hindi" - Founder 1 LinkedIn + Twitter posts on pilot zero testimonial

Sprint 4 — Jul 9 to Jul 22: ComplyShield (Rajasthan + Central)

Engineering: - Rajasthan Coaching Act 2025 rules engine: - Registration application PDF generator - 1 sqm/student calculator with batch-size enforcement - 5-hours/day class limit - 4-installment fee enforcement - Pro-rata refund calculator with UPI payout integration - Counsellor assignment per 200 students - Mental-health helpline auto-footer on parent messages - MoE Central Guidelines 2024: faculty graduation upload + age-gate on enrollment - CCPA Ad Lint: Sarvam-M classifier for banned terms + selection-claim verifier - Inspector Pack: one-click PDF bundle with hash-chain audit trail - Penalty Risk Dashboard: real-time compliance score

GTM: - Continue ads; iterate copy based on Sprint 3 learnings - Convert week-3 trials to paid; aim for 30-40% trial-to-paid conversion - Founder 1 starts asking pilots for referrals

Content: - Video #4: "Rajasthan inspector ko ₹2 lakh penalty se kaise bachan" - Blog post #1: "Why we built compliance-first"

Sprint 5 — Jul 23 to Aug 5: Faculty + Inquiry CRM + Parent Digest

Engineering: - Faculty module: - `faculties`, `contracts`, `batch_assignments`, `ratings` tables - Per-batch revenue attribution; gross margin per teacher - Student-rating micro-survey post-test (1-tap WhatsApp) - Faculty-result analytics: avg student score by faculty - Salary processing with contractual deductions - Inquiry CRM: - WhatsApp lead-capture from Meta/Google ads via Lead Ads API - 60-second auto-greet via WhatsApp template - Demo-class scheduler with calendar integration - Referral-tracking with auto-credit to existing parent - Parent Weekly Digest: - Cron-scheduled Sunday 09:00 IST - Sarvam-M-generated 4-line WhatsApp template - Includes attendance, last 2 scores with peer avg + topper, one "small win", next test - Student Risk Score: - Weighted: attendance 40%, test trend 30%, fee delinquency 15%, contact recency 15% - Threshold breach → auto-flag to counsellor + owner

GTM: - Convert pilots to "Compliance+" tier (₹4,999/mo); price-test the upgrade - **Target: 5 paying customers, ₹15K-30K MRR** - Faculty advisor (IIITA prof) onboarded for XPRIZE submission credibility

Content: - Video #5 + #6: faculty management + retention case studies

Sprint 6 — Aug 6 to Aug 19: Polish + XPRIZE Submission Packaging

Engineering: - Bug fixes from pilot feedback - Performance pass: query plans, RLS index coverage, hot-path caching - Multi-state expansion: UP, Karnataka, Bihar rules added to ComplyShield - 30-day free-trial flow polish; payment upgrade UX - Final security pass: MFA enforcement, secrets rotation, KMS key audit - DeepEval suite green-light

XPRIZE submission: - 500–1000 word narrative (positioning, traction, AI-native moat, category impact) - 3-minute video: founder 1 demos product in English with Hindi subtitles - 0:00–0:30 problem framing - 0:30–1:30 product walkthrough (Onboarding Agent, Smart Dunning, Test Report, ComplyShield) - 1:30–2:30 agent execution logs + Langfuse traces shown - 2:30–3:00 customer testimonial + ARR + traction metrics - Razorpay revenue export - Agent execution log export from Langfuse - 3+ customer testimonial videos - GitHub repo cleanup; README pointing to architecture doc, this document attached

Submission deadline: Aug 17, 2026 (UTC midnight)

10.2 Day 90 success criteria

Criterion	Target	Failure threshold
Paying customers	≥ 5	$< 3 \rightarrow$ call it
MRR	$\geq ₹15K$	$< ₹8K \rightarrow$ call it
Active pilots / trials	≥ 10	$< 5 \rightarrow$ call it
Production AI agents	4 (Onboarding, Test Report, Dunning, Compliance)	$< 3 \rightarrow$ defer XPRIZE
YouTube subscribers	≥ 200	$< 50 \rightarrow$ channel restart
XPRIZE narrative + video	Ready by Aug 17	Missed $\rightarrow$ no submission

Hit ≥ 4 of 6 $\rightarrow$ continue to month 4. Hit $< 3 \rightarrow$ pause and reassess.

11. Team Allocation & Operating Model

11.1 Five-founder structure

Founder	Role	Allocation	Day-90 deliverables
F1	GTM Lead — Hindi sales calls, demos, customer support, pilot onboarding	100%	5–10 paying customers; Hindi demo library
F2	Backend Lead — NestJS, Prisma, Postgres RLS, payments, WhatsApp integration	100%	All core APIs, multi-tenant tested, payments live
F3	Frontend Lead — Next.js 16 web, Expo 52 mobile, dashboards	100%	Owner dashboard, teacher dashboard, parent app shell
F4	AI / Agent Lead — LangGraph, Gemini/Sarvam routing, RAG, evaluation	100%	4 production agents with Langfuse traces
F5	Content + Compliance + XPRIZE Lead — YouTube, blog, compliance research, DPDPA, XPRIZE submission	100%	6 YouTube videos, ComplyShield content base, XPRIZE narrative + video

11.2 Five part-time helpers (10–15 hrs/week each)

- **Helper 1:** QA + manual testing on real coaching workflows
- **Helper 2:** Pilot customer support — Hindi WhatsApp first-line
- **Helper 3:** Content writing — blog posts, social, ad copy
- **Helper 4:** Video editing — YouTube + XPRIZE submission video
- **Helper 5:** Data entry for pilot onboarding — bridge any rough edges

11.3 Faculty advisor

One IITA professor (CS, Software Engineering, or NLP background) as advisor: - 1–2 hours/month - Co-signs XPRIZE submission for credibility - No equity; honorarium ₹0 (relationship-based)

11.4 Operating cadence

- **Daily:** 30-min standup, 09:30 IST
- **Weekly:** Friday retro + demo + metrics review
- **Sprint review:** every 2 weeks
- **Customer call rota:** F1 + one rotating other founder; min 2 calls/week each
- **On-call:** weekly rotation among F2/F3/F4; weekend escalation to F1

12. Unit Economics

12.1 Cost stack (per customer per month, steady state)

Line	Cost
AI inference (Gemini + Sarvam + premium)	₹560
WhatsApp Cloud API (utility messages)	₹30
Hosting (RDS + Fargate + ElastiCache amortized)	₹150
Observability (Langfuse self-host amortized)	₹20
Payment processing (Razorpay 2%)	₹20 (Starter) – ₹100 (Compliance+)
Support (founder time amortized at scale)	₹100
Total COGS	₹880 – ₹960

12.2 Gross margin by tier

Tier	Price	COGS	Gross margin
Starter ₹999	₹999	₹880	12%
Pro ₹2,499	₹2,499	₹920	63%
Compliance+ ₹4,999	₹4,999	₹960	81%

Starter is a loss-leader / acquisition tier. The business model lives on Pro and Compliance+ conversions. Target mix at 100 customers: 30% Starter, 50% Pro, 20% Compliance+ → blended ARPU ₹2,400, blended margin 62%.

12.3 18-month projection (realistic, not optimistic)

Month	Customers	MRR	ARR run-rate
3	5–10	₹15K–25K	₹2L
6	15–25	₹40K–70K	₹6L
12	60–100	₹1.5L–3L	₹25L
18	150–250	₹4L–7L	₹60L
24	300–500	₹8L–15L	₹1.2 Cr

This is a ₹50L–1.5 Cr ARR business, not a unicorn. Bootstrapped, sustainable, defensible via compliance moat.

12.4 Credits stack (year 1)

Source	Estimated value	Effort
Sarvam Startup Program	₹3–5L equivalent	Apply day 1
Anthropic Claude Startup	\$25K–\$100K	Need IIITA prof advisor + apply
Google Cloud / Gemini	\$300 + Founders ~\$2K	Apply day 1
Microsoft for Founders Hub	\$1K Azure + \$2.5K OpenAI	Apply day 1
NVIDIA Inception	\$0 cash, credibility + GPU credits	Instant approval
GitHub Student Pack	\$200 DigitalOcean + Heroku + JetBrains	Verify with <code>.iiita.ac.in</code> email
Total year-1 absorption	~₹6–10L of variable cost	Mostly automated applications

13. XPRIZE Submission Strategy

13.1 Category and scoring

Submitting to: Education & Human Potential (primary), Small Business Services (secondary alignment).

XPRIZE weights three criteria equally:

1. **Business Viability** — real ARR, growth trajectory, defensible moat.
2. **AI-Native Operations** — does the business itself run on AI agents, not just have AI features?
3. **Category Impact** — measurable outcomes for end users (students, coaching owners, parents).

13.2 Narrative angles (one for each criterion)

Business Viability angle: - 5–10 paying customers, ₹15–30K MRR by Day 90 - 18-month path to ₹50L–1 Cr ARR - Defensible compliance moat (Rajasthan Act, CCPA, DPDPA) - Unit economics: 62% blended gross margin

AI-Native Operations angle: - 4 production LangGraph agents handling >70% of operational work - Onboarding Agent: 24-hour activation, 90% automated - Smart Dunning Agent: 5-stage behavioural cadence, RBI-compliant - Test Report Agent: KT + RAG + tool-use, 8 Maurya dimensions evaluated - ComplyShield: continuous rule-monitoring + filing generation - Langfuse execution logs proving end-to-end agent traces

Category Impact angle: - Measurable per-customer outcomes: % faster fee collection, hours saved per teacher, retention lift, compliance violations avoided - Pilot testimonials in Hindi with quantified results - Mental-health duty-of-care alignment (post-Kota crisis)

13.3 Submission package

Deliverable	Owner	Deadline
GitHub repo public + README	F2 + F4	Aug 10
Architecture document (this doc) attached as appendix	F5	Aug 10
500–1000 word narrative	F5	Aug 12
3-minute video	F1 + F5 + Helper 4	Aug 14
Razorpay revenue export	F2	Aug 14
Langfuse agent execution log export	F4	Aug 14
3+ customer testimonial videos	F1 + F5	Aug 14
Final polish + submit	F5	Aug 17

13.4 What submission cannot fix

This is lottery economics. The marginal cost is ~60 hours of submission packaging on top of work being done regardless. Even at 0.5% win probability the EV justifies submitting. But:

- The real business goal is sustainable ₹50L–1 Cr ARR, not XPRIZE.
- Do NOT warp the product to please judges over paying customers.
- Use the deadline as a forcing function for shipping, not as a north star.

14. Risk Register

14.1 Top 10 risks ranked by probability × severity

#	Risk	Probability	Severity	Mitigation
1	No customer pickup in 90 days	60%	High	Concierge pilot Day 0 from family network; Hindi founder calls; 30-day free trial
2	Meta WhatsApp Business AI commoditizes our channel	90% (already live)	Medium	Position as <i>complement</i> to WhatsApp AI, not replacement; differentiate on deep coaching-specific data
3	Classplus discount-bombs to ₹999	35%	High	Compliance moat (they don't have it); per-student pricing optionality
4	Pilot customer churn within 60 days	50%	High	Concierge onboarding; success-manager touch by F1; weekly business review
5	Founder burnout / team dissolution	40%	Critical	Clear roles; faculty advisor; weekly retro; explicit "call it" thresholds
6	DPDPA enforcement misstep → penalty	10% (until May 2027), 30% after	Critical	10-item MVP compliance checklist non-negotiable; cyber insurance Q4 2026

7	LLM cost overrun	25%	Medium	Per-tenant circuit breakers; aggressive Gemini Flash prompt caching; Sarvam credits
8	OCR quality fails on field conditions	50%	Medium	Sarvam Vision primary + Gemini Pro fallback; teacher HITL approval blocks bad output
9	Razorpay subscription disputes / chargebacks	20%	Medium	Pro-rata refund logic; clear T&Cs; never auto-debit beyond plan
10	XPRIZE submission misses deadline	30%	Low (business doesn't depend on it)	F5 owns; Aug 14 internal deadline 3 days before public deadline

14.2 Walk-away triggers

If any of these hit by Day 90, pause and reassess:

- 0 paying customers by Day 60
- < ₹8K MRR by Day 90
- All 5 pilot customers churn before Day 60
- A founder leaves
- Critical DPDPA violation discovered before Sept 2026 ship

15. Open Questions for Peer Review

We are sharing this document with Gemini, DeepSeek, and other AI reviewers for peer critique. The following are open questions where we'd particularly value pushback:

LangGraph vs Claude Agent SDK for the orchestration layer — we chose LangGraph for durable execution and PostgresSaver checkpointing. Would Claude Agent SDK's subagent isolation model be a better fit given our mixed-model strategy?

Agent service language: Python (LangGraph) introduces a second runtime alongside Node (NestJS). Would a TypeScript-only stack with `@langchain/langgraph-sdk-js` reduce ops complexity at acceptable feature cost?

Vector store: pgvector + pg_search in one Postgres simplifies ops but pgvector hits a wall past ~50M chunks. Should we plan a Qdrant migration trigger from day one, or defer until we see the wall?

Multi-tenancy enforcement: RLS via Prisma extension feels brittle (one missed `set_config` and isolation breaks). Should we layer in WorkOS-style permission boundaries as a defense-in-depth, or trust the extension + RLS forced policy?

DigiLocker VPC fallback: how robust is the DigiLocker OAuth flow in 2026 for parents who don't have Aadhaar-linked DigiLocker yet? Should we plan a phone-OTP-only fallback that meets DPDPA Rule 10 specifically?

Crypto-shredding for right-to-erasure: per-tenant DEK envelope encryption is clean but operationally heavy (key rotation, backup, recovery). Is application-level pgcrypto + cascading DELETE sufficient for DPDPA's 90-day SLA?

Smart Dunning ethical edge: the Dutch field experiment shows 23% repayment lift from AI dunning, but the psychological-impact study warns of "stigma reduction" framing. Are we drawing the right line on tone escalation in stages 3–5?

Sarvam concentration risk: we depend on Sarvam for translate + STT + TTS + OCR. They're well-funded (\$1.5B valuation, HCLTech-led Series B) but still pre-IPO. Should we self-host AI4Bharat IndicTrans2 + Whisper as a permanent fallback, even at higher ops cost?

Compliance wedge sustainability: the Rajasthan / CCPA / DPDPA stack is real today. But what's the 3-year half-life of regulatory pain? Will competitors (Classplus, Teachmint) ship the same features within 12–18 months and erase our moat?

GTM realism: research says no Indian SMB SaaS at ₹1.5K–4K/month ever scaled purely online-only. We're betting on founder-led Hindi Zoom demos + Click-to-WhatsApp ads + freemium. Is there a channel we're missing?

XPRIZE category fit: Education & Human Potential vs Small Business Services — which category is the right submission slot given that our buyer is the coaching owner (small business) but our impact is on students?

Pricing tier sequencing: should we launch only Pro ₹2,499 in Sprint 1–3 and add Starter ₹999 + Compliance+ ₹4,999 in Sprint 4–6, or all three from Sprint 1? Starter is a margin loss-leader; launching it too early may anchor pricing low.

16. Research Bibliography

16.1 AI agents in education

- Maurya et al., *Unifying AI Tutor Evaluation* (arXiv 2412.09416)
- Maurya et al., *BEA 2025 Shared Task on Pedagogical Ability Assessment* (arXiv 2507.10579)
- Wang et al., *TutorLLM: Knowledge Tracing + RAG* (arXiv 2502.15709)
- Stanford SCALE, *Tutor CoPilot* (arXiv 2410.03017)
- Kestin et al., *AI Tutor RCT* (Nature Scientific Reports 2025) — nature.com/articles/s41598-025-97652-6
- Khan Academy, *Khanmigo* — khanmigo.ai
- CMU, *AI Math Tutor Helping Millions* — cmu.edu/research-office/research-impacts/ai-math-tutor.html
- LPITutor (PeerJ CS 2025) — ncbi.nlm.nih.gov/pmc/articles/PMC12453719/

16.2 LangGraph & agent orchestration

- Anthropic, *Building Effective Agents* (Dec 2024) — anthropic.com/research/building-effective-agents
- Anthropic, *Multi-agent Research System* — anthropic.com/engineering/multi-agent-research-system
- Anthropic, *Claude Agent SDK* — anthropic.com/engineering/building-agents-with-the-claude-agent-sdk
- LangGraph docs — docs.langchain.com/oss/python/langgraph/memory
- LangGraph case study, Remote — langchain.com/blog/customers-remote
- LangGraph vs CrewAI vs AutoGen 2025 — latenode.com/blog/.../langgraph-vs-autogen-vs-crewai...
- DataCamp comparison — datacamp.com/tutorial/crewai-vs-langgraph-vs-autogen

16.3 Multi-tenant SaaS

- PlanetScale, *Approaches to tenancy in Postgres* — planetscale.com/blog/approaches-to-tenancy-in-postgres
- Crunchy Data, *RLS for Tenants in Postgres* — crunchydata.com/blog/row-level-security-for-tenants-in-postgres
- Prisma row-level-security extension — github.com/prisma/prisma-client-extensions/tree/main/row-level-security

- Notion DB sharding case study — talent500.com/blog/notion-postgresql-database-sharding/
- WorkOS multi-tenant permissions — workos.com/blog/multi-tenant-permissions-slack-notion-linear
- Vercel multi-tenant SaaS docs — vercel.com/docs/multi-tenant

16.4 Vector stores & RAG

- pgvector vs Pinecone vs Qdrant vs Weaviate — dev.to/kencho/...vector-database-performance-compared...
- AWS pgvector indexing — aws.amazon.com/blogs/database/optimize-generative-ai-applications-with-pgvector-indexing...
- ParadeDB hybrid search — paradedb.com/blog/hybrid-search-in-postgresql-the-missing-manual
- Notion two-years vector search — notion.com/blog/two-years-of-vector-search-at-notion
- OptixStack hybrid search + reranking playbook — optixstack.com/rag-reliability/hybrid-search-reranking-playbook

16.5 Indic AI

- Sarvam-1 / OpenHathi — sarvam.ai/blogs/sarvam-1
- Sarvam-M (May 2025) — sarvam.ai/blogs/sarvam-m
- Sarvam Vision (OCR) — sarvam.ai/blogs/Sarvam-vision
- Sarvam Startup Program — sarvam.ai/startup-program
- AI4Bharat — ai4bharat.iitm.ac.in; IndicTrans2 — github.com/AI4Bharat/IndicTrans2
- Krutrim LLM paper (arXiv 2502.09642)
- COMI-LINGUA Hinglish dataset (arXiv 2503.21670)
- IndicParam benchmark — medium.com/@vbsowmya/indicparam...
- Vyakyarth-1 — ai-labs.olakrutrim.com/models/Vyakyarth-1-Indic-Embedding

16.6 WhatsApp Business

- Meta webhook reference — developers.facebook.com/documentation/business-messaging/whatsapp/webhooks/reference/messages/
- WhatsApp template messages — developers.facebook.com/documentation/business-messaging/whatsapp/templates/overview
- WhatsApp Cloud API 2026 — medium.com/@aktyagihp/whatsapp-cloud-api-integration-in-2026...

- Meta Business AI India launch (May 2026) — about.fb.com/news/2026/05/introducing-business-ai-on-whatsapp-for-small-businesses-in-india/
- Webxion: WhatsApp fee alerts for coaching — webxion.com/how-coaching-institutes-send-fee-alerts-on-whatsapp-business-platform/

16.7 DPDPA & regulatory

- Rule 10 (Children's data & VPC) — dpdpa.com/dpdparules/rule10.html
- Rule 12 (SDF + localisation) — dpdpa.com/dpdparules/rule12.html
- Rule 14 (Rights & SLA) — dpdpa.com/dpdparules/rule14.html
- PIB official PDF — static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc20251117695301.pdf
- Grant Thornton brochure — granthornton.in/.../dpdpa-rules-detailed-brochure_final-25th-november-2025-1.pdf
- EY DPDPA guide — ey.com/en_in/insights/cybersecurity/decoding-the-digital-personal-data-protection-act-2023
- Penalties guide — dpdpa.com/blogs/dpdpa_penalties_explained_50_crore_250_crore_fines.html
- Securiti cross-border — securiti.ai/cross-border-data-transfer-requirements-under-india-dpdpa/
- DigiLocker framework — dpo-india.com/Resources/.../Digital-Locker-Tecnology-Framework-Verifiable-Parental-Consent-India's-
- Hogan Lovells IT Rules 2026 — hoganlovells.com/en/publications/india-introduces-mandatory-labelling-for-ai-and-3hour-takedown-for
- PowerSchool breach lessons — mcmillan.ca/insights/publications/lessons-learned-from-the-powerschool-breach/

16.8 Coaching center operations

- IMARC India Coaching Market Report — imarcgroup.com/india-coaching-institutes-market
- NSS 2025 PIB release — pib.gov.in/PressReleasePage.aspx?PRID=2160863
- Rajasthan Coaching Centres Act 2025 (PRS brief) — prsindia.org/files/bills_acts/bills_states/rajasthan/2025/Legislative_Brief_Rajasthan_Coaching_Cent
- MoE Coaching Guidelines 2024 — education.gov.in/sites/upload_files/mhrd/files/Guideliens_Coaching_Centres_en.pdf
- PIB CCPA penalties on coaching — pib.gov.in/PressReleaseIframePage.aspx?PRID=2073013
- ThePrint FIITJEE crisis — theprint.in/india/education/fiitjee-crisis-angry-parents-allege-big-fraud-teachers-who-quit-join-ri

- SAGE Kota suicide study (Meena 2025) — journals.sagepub.com/doi/10.1177/09710973251356797
- Microsoft Research Physics Wallah Alakh AI — microsoft.com/en-us/research/blog/microsoft-research-and-physics-wallah-team-up-to-enhance-ai-based
- ClassPlus / Inc42 pivot coverage — inc42.com/features/classplus-flips-its-edtech-playbook/

16.9 Behavioural economics & dunning

- PNAS, *Behavioural Nudges Prevent Loan Delinquencies* — ncbi.nlm.nih.gov/pmc/articles/PMC11789030/
- Netherlands AI debt collection field experiment — ncbankruptcyexpert.com/sites/default/files/2024-07/artificial_intelligence_and_debt_collection_evid
- AI in Debt Collection: Psychological Impact (arXiv 2602.00050)
- RBI FREE-AI Framework — lexology.com/library/detail.aspx?g=cdd93d6c-fd28-4c12-ac23-33d7820439ab
- WhatsApp notifications & micro-conversions in fintech — chatarchitect.com/news/fintech-meets-messaging-whatsapp-based-notifications-and-micro-conversions

16.10 Evaluation & safety

- LLM-as-Judge Survey (arXiv 2411.15594)
- Empirical Study of LLM-as-Judge (arXiv 2506.13639)
- Implicit Grading Bias in LLMs (arXiv 2603.18765)
- FairAIED (arXiv 2407.18745)
- SRPG Privacy Guard for Ed-MAS (arXiv 2512.03694)
- HITL non-negotiable for safety-critical AI — itrevolution.com/articles/human-in-the-loop-is-non-negotiable-leading-ai-adoption-in-safety-critical
- DeepEval changelog 2025 — deepeval.com/changelog/changelog-2025

End of document. Version 1.0. To be revised based on peer review and pilot feedback by Sprint 3 close (July 8, 2026).